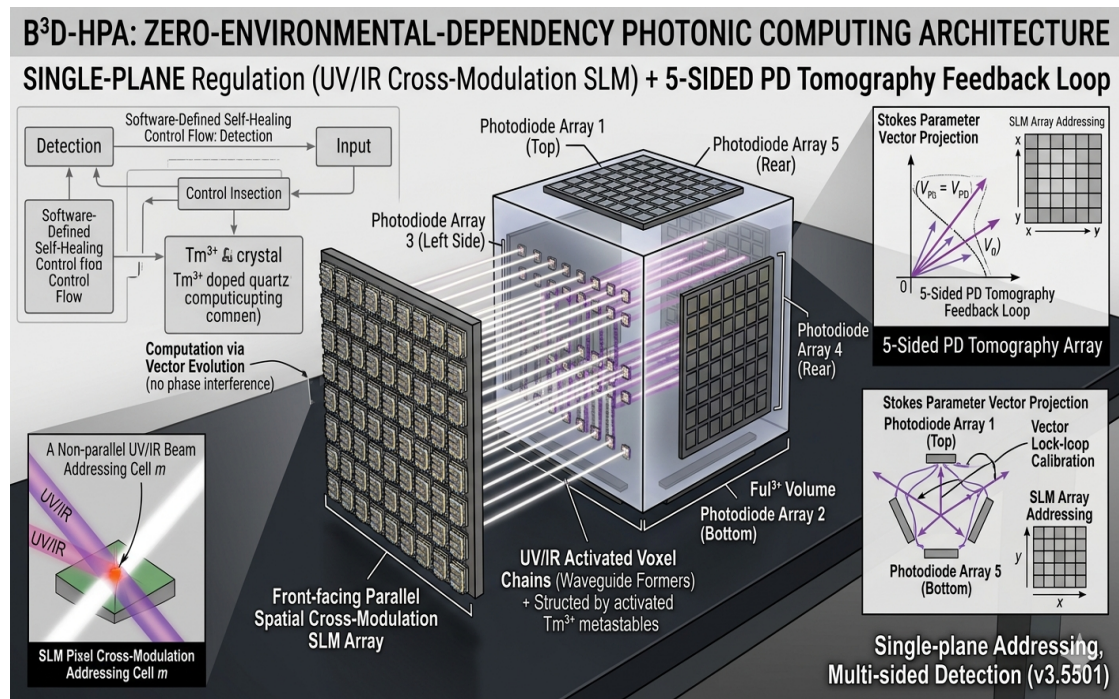




In the B<sup>3</sup>D-HPV paradigm, matrix inversion is no longer a high-complexity silicon computation, but a physical collapse of the Hermitian Adjoint operator. By leveraging the geometric nature of polarization, we achieve near-zero latency inversion through optical conjugation.

In the B<sup>3</sup>D-HPV (Physics-based Volumetric Logic) paradigm, we move away from the high-complexity iterative processes of silicon-based logic. Instead, we treat mathematical operations as geometric projections and physical state collapses.

### 1. Matrix Inversion: The "Physical Collapse"

In traditional computing, inverting a matrix  $M$  requires  $O(n^3)$  complexity (e.g., Gaussian elimination). In our photonic architecture, we leverage the Unitary nature of polarized optical flow.

**The Logic:** If a transformation matrix  $M$  is represented by a series of lossless polarization rotations (Unitary transformations), then its inverse  $M^{-1}$  is simply its Hermitian Adjoint  $M^\dagger$  (the conjugate transpose).

**The Implementation:** In B<sup>3</sup>D-HPV, "calculating" the inverse is not an arithmetic operation, but a Symmetry Transformation. By reversing the polarization state or utilizing the geometric reciprocity of the quartz lattice, the inversion occurs as a near-zero latency physical collapse.<sup>6</sup>

**The Advantage:** We achieve  $O(1)$  complexity. The answer is not "computed"; it is

"revealed" by the physical symmetry of the optical field.

## 2. The Polarization Adder (POL\_ADD)

Photonic addition is naturally handled by the Principle of Superposition.

Principle: When two incoherent light fields  $I_1$  and  $I_2$  are combined into the same spatial mode (e.g., through a Beam Combiner), the resulting intensity is a direct summation.

Vector Mapping: By mapping data values to the intensity or the amplitude of polarized wave-fronts, the hardware performs massive parallel addition simply by letting the light paths merge within the 3D quartz structure.

## 3. The Polarization Subtractor (POL\_SUB)

Subtraction is the historical "Achilles' heel" of incoherent optical computing, as light intensity cannot be negative. B<sup>3</sup>D-HPV solves this via Geometric Projection Mapping.

The Mechanism: Instead of trying to "cancel" photons (which requires unstable phase interference), we use Polarization Orthogonality.

Process:

Encoding: Map the minuend (A) to the Horizontal axis (0°) and the subtrahend (B) to the Vertical axis (90°).

Rotation: The SLM executes a POL\_TRANS instruction, rotating the composite polarization vector by a specific angle  $\theta$ .

Projection: We use a Polarization Sensitive Detector (or a PBS) to extract the projected components. By measuring the difference in intensity between the two orthogonal projections, we physically extract the value  $A - B$ .

Result: This is a Robust Subtraction. Unlike phase-based destructive interference, it is immune to thermal phase drift because it relies on the rigid geometric orientation of the polarization states.

Summary: Geometry vs. Arithmetic

Addition — Silicon (Digital): Gates & Latency; B<sup>3</sup>D-HPV (Geometric): Superposition ( $O(1)$ )

Subtraction — Silicon (Digital): Two's Complement; B<sup>3</sup>D-HPV (Geometric): Orthogonal Projection

Inversion — Silicon (Digital): Iterative Loops ( $O(n^3)$ ); B<sup>3</sup>D-HPV (Geometric): Hermitian Collapse ( $O(1)$ )

By defining these as Physical Mapping Instructions (PDMM), we turn the quartz lattice into a high-dimensional geometric computer where the "logic" is simply the evolution of the light field's geometry.

Key Updates & Version Notes

This version incorporates the finalized system-level schematic, completing the closed-loop design from conceptual logic to engineering implementation, and adds the software-programmable reference voltage ( $V_{ref}$ ) of PD front-end comparators to realize dynamic adjustment of AGI computing sensitivity, further improving the software-defined capability of the architecture.

### 1. Core System: Single-Plane UV/IR Spatial Cross-Modulation SLM + 5-Sided PD Array

The design diagram confirms the single-plane, side-pumping-free architecture, using only a front-facing UV/IR mixed parallel SLM array to achieve:

- 3D voxel addressing: Non-parallel UV/IR beams intersect at a focal point to form an energy threshold, activating  $Tm^{3+}$  metastable traps precisely at the (x,y,z) voxel without path damage.
- Polarization operator writing: The IR beam induces local birefringence at the activated voxel, directly implementing polarization-based logic gates.
- 5-sided PD tomography: Top/bottom/left/right/rear photodiode arrays capture scattered and transmitted light for full-volume topology reconstruction and runtime calibration, with the 5-channel differential PD core eliminating common-mode noise from intensity superposition; the front-end high-speed comparators are equipped with software-programmable  $V_{ref}$  to support dynamic AGI sensitivity adjustment.

### 2. Polarization Incoherent Computing Paradigm

Abandons phase interference for high robustness, with the core advantage of spatial topology replacing temporal phase:

- Logic states are mapped to Poincaré sphere coordinates, and polarization evolution follows unitary matrix rules with inverse computation via physical conjugate transposition.
- Computation is realized via UV/IR-induced local anisotropy instead of phase superposition, doubling the initial parallelism compared with single-phase schemes.
- Maintains high SNR even with thermal drift through 5-channel differential PD noise suppression and programmable  $V_{ref}$  threshold calibration, eliminating the need for precision environmental control and expensive anti-vibration platforms.

### 3. Multi-Method Virtual Waveguide Writing

Multiple refractive index gradient induction strategies are supported, selectable or combinable, all compatible with unitary polarization manifold computing:

4. UV/IR cross-modulation (native core method)
5. Photothermal-assisted writing
6. Photo-induced refractive index (PRI) enhancement
7. Multi-wavelength IR cooperative confinement
8. Voxel-level array superposition

### 9. Software-Defined Layer: Topology Equivalence & Self-Healing Control

The core paradigm of "flexible logical topology" is formalized with three key algorithms,

supporting real-time calibration for polarization evolution stability and software-controlled AGI sensitivity adjustment:

- Optical Homing: SLM-to-voxel dynamic mapping via blind scanning and trilateration.
- Self-Organizing Routing: Reinforcement learning-based path formation via PD feedback with programmable Vref threshold.
- Vector Lock-loop: Real-time polarization correction via Stokes parameter monitoring, synchronously calibrating the comparator Vref to match AGI task requirements.

#### 5. Photonics HLS: From High-Level Language to Physical Collapse

Defines the complete compilation flow, directly mapping PDMM unitary polarization operators to Optical ISA, and integrating AGI sensitivity adjustment into the optical instruction set:

6. Human layer: Logical abstraction and Optical ISA definition (compatible with POL/TM series operators, adding sensitivity adjustment instructions).
7. Silicon layer: Topological sketching and linear mapping, converting sensitivity requirements into comparator Vref calibration parameters.
8. SLM layer: Parallel execution and runtime verification, synchronizing Vref adjustment with polarization operator implantation.

- Supports both static curing (photonics FPGA) and streaming trigger (dynamic processor) modes, with a recommended "static skeleton + dynamic fine-tuning" hybrid, and dynamic Vref adjustment is compatible with both operating modes.

#### 6. Core Value & Future Directions

- Simplified, robust, software-defined, and mass-production-ready, with  $O(1)$  time complexity for unitary matrix inverse computation and software-programmable AGI computing sensitivity via comparator Vref adjustment.
- Future work includes ISA development with complete sensitivity adjustment instruction sets, millisecond rerouting, prototype integration, waveguide writing method optimization for polarization computing, and dynamic sensitivity matching algorithm optimization for complex AGI tasks.

B<sup>3</sup>D-HPV Lanthanide Quartz Photonic Computing Chip Supplementary Paper Memorandum: Optical Field Control and Software-Defined System for Polarization Incoherent Computing

#### I. New Additions to the Core System: 3D Voxel Addressing and Polarization Incoherent Computing Based on Single-Plane Regulation

##### 1. Dual-Band Cross-Modulation Architecture Without Side Pumping

Abandoning the traditional side pumping structure, this system achieves 3D voxel

addressing and polarization operator writing solely through a front-side parallel infrared/ultraviolet hybrid array spatial cross laser modulation SLM, realizing path-error-free fixed-point activation via the spatial intersection of dual-band non-parallel optical fields:

- Ultraviolet addressing channel: Dual-wavelength ultraviolet optical fields are incident at non-parallel angles, forming an energy density threshold only at the intersection points to precisely excite  $\text{Tm}^{3+}$  metastable traps, lock the (x,y,z) 3D voxel positions, and eliminate the need for additional side optical paths;
- Infrared computing channel: 1550 nm communication band infrared optical fields are incident synchronously, undergoing polarization modulation with activated  $\text{Tm}^{3+}$  ions at the intersection points to induce local birefringence/dichroism and directly realize physical implantation of polarization operators including unitary conjugate transposition inverse;
- Core advantages: Achieve the decoupling of addressing and computing through the geometric characteristics of optical fields, fundamentally eliminate path errors, simplify the optical path structure, and realize monolithic integration simultaneously, laying the physical foundation for spatial topology-based computing and software-controlled dynamic sensitivity adjustment.

## 2. Polarization Incoherent Computing Paradigm

Abandoning the reliance on precise phase interference, this paradigm constructs a highly robust computing logic with the polarization evolution of vector optical fields and the statistical response of medium anisotropy as the core, realizing the paradigm shift of spatial topology replacing temporal phase, and adding software-programmable sensitivity adjustment based on comparator  $V_{\text{ref}}$ :

- Map logical states to Poincaré sphere coordinates, with the spatial evolution of polarization states serving as the logical carrier; all polarization evolution strictly follows unitary matrix rules, and inverse matrix computation is directly realized by physical conjugate transposition of polarization states with  $O(1)$  time complexity;
- Complete computing through local birefringence/dichroism induced by UV/IR cross modulation, rather than optical wave phase superposition; this polarization scheme inherently doubles the initial parallelism compared with single-phase schemes, with computing completed by spatial geometric modulation instead of time-domain phase locking;
- Incoherent superposition core mechanism: 5-channel differential PD array eliminates common-mode noise from intensity superposition, and the front-end comparator with software-programmable  $V_{\text{ref}}$  enables polarization projection to achieve logical judgment with the same precision as phase interference, and supports dynamic adjustment of judgment threshold according to AGI task requirements;
- As long as the polarization basis vectors remain stable, the signal-to-noise ratio of computing maintains an extremely high level even if the phase drifts with temperature;

the programmable Vref can real-timely calibrate the judgment threshold to compensate for environmental drift, breaking away from the dependence on precise temperature control and expensive anti-vibration platforms.

## II. Five-Side PD Array: Polarization Tomography and Full-Link Closed Loop

### 1. Functional Upgrade: From Signal Receiver to Polarization Tomography Scanner

The five-side PD array (upper, lower, left, right sides + front side) is no longer a mere signal acquisition terminal, but forms a polarization tomography system to realize 3D reconstruction of the polarization evolution of optical fields inside the quartz cube, with its 5-channel differential PD core being the physical foundation for noise suppression and high-precision computing, and the front-end comparators equipped with software-programmable Vref to realize dynamic AGI sensitivity adjustment:

- The upper, lower, left and right PDs capture the Stokes parameters of side scattered light generated by polarization plane rotation; the front PD captures the main signal flow; the photocurrent output of all PDs is sent to the high-speed comparator for differential judgment, converting analog light intensity signals into digital polarization logic states;
- The silicon-based controller inversely solves the computing result of the input vector  $X$  processed by the polarization conversion matrix  $M$  of the quartz cube through the linear matrix operation  $Y = M \cdot X$ , without the need to know the specific optical path;
- The 5-channel differential PD array matches the  $Tm^{3+}$  1.4~2.0 $\mu m$  emission band, achieving a common-mode rejection ratio (CMRR)  $\geq 65$  dB, providing high SNR guarantee for unitary conjugate transposition inverse computation and spatial topology-based logical judgment;
- The front-end high-speed comparators integrate a software-programmable reference voltage (Vref) module, which is directly controlled by the silicon-based controller: the software layer issues Vref calibration instructions according to different AGI task sensitivity requirements (high sensitivity for high-dimensional tensor computation, low sensitivity for large model coarse-grained inference), and the hardware layer real-timely adjusts the judgment threshold of the comparator, realizing the flexible matching between physical polarization judgment and algorithmic dynamic demand without modifying the optical topology.

### 2. Real-Time Self-Calibration Closed Loop

During the logic writing process, the five-side PD array synchronously collects the energy level projection distribution of  $Tm^{3+}$  ion luminescence, realizing real-time monitoring of "write as seen" and ensuring the accuracy of polarization operator implantation; the closed-loop system synchronously calibrates the comparator Vref to maintain the consistency of AGI computing sensitivity while correcting writing deviations:

- If the  $Tm^{3+}$  excitation at a certain voxel position is insufficient, the silicon-based system

immediately fine-tunes the infrared power of the corresponding pixel of the SLM; if the polarization state drift causes the PD output to deviate from the target range, the system synchronously adjusts the comparator  $V_{ref}$  to compensate for the drift, ensuring the stability of logical judgment and sensitivity;

- Correct writing deviations in real time, solve the pain points of "uncertain writing and computing precision drift" in photonic computing, and maintain the stability of unitary polarization evolution, conjugate transposition inverse computation and AGI dynamic sensitivity adjustment;
- The  $V_{ref}$  calibration data is fused with the Stokes parameter feedback of the Vector Lock-loop, forming a unified closed-loop control table, which realizes the joint optimization of polarization state correction and sensitivity adjustment, and improves the overall operational reliability of the architecture.

### 3. Technical Supplement: The Logic Hardening Layer (LHL)

Subsystem: PD-Differential Decision Matrix

#### 1. What is the Comparator?

In this architecture, the comparator is a high-speed analog-to-digital decision primitive. Unlike a multi-bit Analog-to-Digital Converter (ADC) that quantizes signals into complex, resource-heavy data packets, the comparator performs a singular physical decision. It acts as the "Event Horizon" where the continuous, overlapping Stokes vectors of the quartz medium are collapsed into discrete binary states (0 or 1).

#### 2. Why Use a Comparator? (The Anti-Inversion Strategy)

The primary challenge in non-coherent polarization computing is Computational Inversion—a state where the energy and latency required for a digital processor to interpret a raw analog signal exceed the utility of the optical calculation itself.

- Zero-Latency Hardening: It eliminates the need for CPU-intensive sampling and Digital Signal Processing (DSP). The decision occurs at the speed of electron drift within the silicon, bypassing software instruction cycles.
- Noise Immunity (Hysteresis): By implementing a Schmitt-trigger configuration, the comparator ignores the micro-fluctuations and thermal noise inherent in non-coherent light environments, ensuring only significant polarization state shifts trigger a logical output.
- Protocol Alignment: Humans and digital systems cannot natively interact with a continuous wave. The comparator translates the "Nature's Rhythm" of the quartz into the "Human Logic" of the system bus.

#### 3. Theory of Operation: Geometric Thresholding

The operation leverages a 5-Channel Differential Photo-Diode (PD) Array to perform spatial-to-logical mapping.

#### A. Input Vector Mapping

The PD array captures the Stokes parameters ( $S_1$ ,  $S_2$ ,  $S_3$ ) via differential intensity measurements:

$S_1 = I_H - I_V$  (Horizontal vs. Vertical)

$S_2 = I_{45} - I_{135}$  (Diagonal vs. Anti-diagonal)

$S_3 = I_R - I_L$  (Right vs. Left Circular)

These differential signals are fed directly into the comparator's inputs. To ensure robustness against source fluctuations, we utilize a Dynamic Reference ( $V_{ref}$ ) derived from the total intensity ( $S_0$ ).

The logic output is determined as:

$V_{out} = \text{High (1) if } V_{sig} > V_{ref} \pm \Delta H$

$V_{out} = \text{Low (0) if } V_{sig} < V_{ref} \mp \Delta H$

where  $\Delta H$  represents the hysteresis window.

#### B. Unitary Operator Validation & Real-time Monitoring

The fundamental integrity of the optical computation is guaranteed by the Unitarity of the operator matrices implemented by the quartz crystal and the SLM (Spatial Light Modulator). In the EDA backend, all logic masks are validated against the unitary condition to ensure zero information loss during the evolution phase.

This unitarity is continuously verified in real time using the  $S_i$  data from the PD array. Any significant drift in these values from the expected state—indicating physical displacement or thermal expansion—triggers an automated "Logic Realignment" procedure to recalibrate the optical path.

#### C. Physical State Collapse

As the polarization state rotates on the Poincaré Sphere due to the quartz crystal's geometric phase evolution, its projection on the measurement axes changes. When the vector crosses the "Equatorial Decision Plane" defined by the comparator's threshold, the hardware forces an instantaneous logical flip. The 5-channel differential design ensures these projections are captured with high contrast and stability, even in low-coherence conditions.

#### D. The Result: Causal-Chain Constraint ( $C^3$ )

The output is a clean, jitter-free square wave. This pulse represents a Causal-Chain Constraint ( $C^3$ )—a physical proof that the quartz logic has reached a specific state. This output is ready for immediate consumption by a general-purpose processor's GPIO or for triggering downstream SLM control loops without overhead.



## Engineering Note:

The comparator is the Filter of Reality. By discarding the "process" of the quartz evolution and only reporting the "result," we achieve a high-fidelity interaction that remains robust even in non-coherent, high-entropy environments.

"Quartz calculates via evolution; the comparator decides via projection. Without the latter, the former remains silent to the human world."

## Disclaimer:

The proposed non-coherent geometric polarization computing method is a pioneering exploration in an uncharted research field. Any errors or inaccuracies in the method are purely accidental, and no liability shall be assumed for any consequences arising therefrom.

## III. Multi-Path Writing Implementation Methods for Virtual Waveguides

In this architecture, virtual waveguides are dynamic optical field confinement structures without physical etching. Their core is to realize the directional guidance of infrared computing light by inducing local refractive index gradients/anisotropic distributions inside the quartz cube through optical field modulation. Compatible with the SLM single-plane regulation and five-side PD closed-loop system with programmable  $V_{ref}$ , they have a variety of selectable/combinable implementation methods, each forming differentiated adaptation characteristics around the "refractive index gradient induction method, polarization sensitivity, stability/dynamics", and all support the physical implementation of unitary polarization manifold operators and the matching of AGI dynamic sensitivity requirements:

1. Local birefringence induction method via UV/IR cross modulation (native core method)  
UV+IR spatial cross optical fields output by the SLM activate  $Tm^{3+}$  metastable states at specified voxel points to induce local dielectric tensor distortion of the quartz lattice, forming a polarization-sensitive refractive index gradient that only exerts directional light guidance on infrared computing light with specific polarization states and has no confinement on non-target polarization light; writing and addressing are completed synchronously without additional optical paths, serving as the core method with the highest engineering feasibility, and can directly implant PDMM core operators such as POL\_TRANS and unitary conjugate transposition inverse POL\_INV in voxels; the polarization-sensitive characteristic of the waveguide matches the programmable  $V_{ref}$  of the PD comparator, realizing the precise matching of optical layer polarization evolution and electrical layer judgment sensitivity for AGI fine-grained computing tasks.

2. Photothermal effect-assisted method

Utilize the photothermal effect of ultraviolet light to form micro-scale temperature gradients on the specified paths inside the quartz cube, generate continuous refractive

index gradients in the path area relying on the thermo-optical coefficient of quartz, and realize full-polarization light guidance confinement for infrared light; the power distribution of ultraviolet light can be regulated by the SLM to precisely control the range and gradient value of thermal gradients, adapting to scenarios requiring wide polarization compatibility, with slight thermal crosstalk that needs real-time fine-tuning in conjunction with the five-side PD array to maintain the stability of polarization evolution; the full-polarization confinement characteristic is suitable for AGI coarse-grained inference tasks with low sensitivity requirements, and the comparator  $V_{ref}$  can be set to a high threshold to filter thermal crosstalk noise and improve computing robustness.

### 3. Photo-induced refractive index change (PRI) enhancement method

Perform trace photosensitive doping (e.g., germanium, tin) on  $Tm^{3+}$ -doped quartz, utilize the photo-induced refractive index change effect of ultraviolet light to make the quartz matrix on the cross optical field path produce permanent/semi-permanent refractive index enhancement, forming stable virtual waveguides; suitable for photonic FPGA scenarios requiring long-term curing of logical topologies and fixed unitary polarization operator implantation, with relatively high reconstruction difficulty after writing that requires higher-power ultraviolet light for erasure; the stable waveguide characteristic is matched with fixed  $V_{ref}$  threshold calibration, suitable for AGI fixed inference tasks with constant sensitivity requirements, and the  $V_{ref}$  parameter is solidified in the silicon-based controller to ensure long-term computing consistency.

### 4. Multi-wavelength infrared cooperative confinement method

Without ultraviolet addressing light, directly output multiple phase-matched infrared light beams through the SLM to form interference-type refractive index gradients by utilizing the transient refractive index change of the optical Kerr effect, and quickly construct dynamic virtual waveguides; featuring an extremely fast writing speed, it is suitable for dynamic processor scenarios with "streaming trigger" and real-time adjustment of polarization computing paths, requiring slightly higher infrared light power, with a short waveguide maintenance time that needs continuous optical field refreshment; the dynamic reconfigurability of the waveguide is fully compatible with the real-time  $V_{ref}$  adjustment of the comparator, realizing the synchronous update of optical layer topology and electrical layer sensitivity for AGI online learning tasks with real-time changing algorithm requirements.

### 5. Voxel-level array superposition method

Decompose virtual waveguides into multiple continuous 3D voxel points, activate voxel points one by one through the SLM parallel array to form a discrete refractive index gradient array, and realize continuous path light guidance through directional scattering/refraction of infrared light between adjacent voxel points; adapting to the construction of virtual waveguides with complex 3D topologies, it can flexibly reconstruct paths by turning off individual voxel points, serving as the core method for realizing "logical rerouting" and highly compatible with self-healing algorithms, supporting dynamic adjustment of spatial topology-based polarization computing; the voxel-level fine-grained control of the waveguide is matched with the precise  $V_{ref}$  adjustment of the comparator, realizing the partitioned sensitivity control of the chip, and supporting multi-engine parallel computing with different sensitivity requirements on a single chip for complex

AGI multi-task fusion scenarios.

A combined strategy can be adopted in actual implementation, such as curing the core topology and unitary polarization operators via UV/IR cross modulation and fine-tuning dynamic paths via the photothermal effect, which balances the stability and reconfigurability of the system; the corresponding Vref adjustment strategy is formulated for different waveguide combination methods, realizing the joint optimization of optical layer waveguide characteristics and electrical layer judgment sensitivity, and conforming to the core design requirements of flexible topology, self-healing and software-defined AGI sensitivity.

#### IV. Software Control Layer: Topological Equivalence and Self-Healing Reconfigurable Photonic Computing

##### 1. Core Paradigm: Logical Topological Equivalence

The SLM array writes not fixed addresses but flexible logical topologies, the core of which is spatial topology replacing temporal phase, and the logical transformation function is the core of polarization computing rather than specific optical paths; the paradigm is extended to the electrical layer, taking AGI computing sensitivity as a software-defined parameter and realizing the flexible matching of physical layer judgment threshold and algorithm layer demand through programmable Vref:

- As long as the logical transformation function  $f(x)$  between input and output is correct, the specific optical paths of light inside the cube (straight line, detour, multi-voxel relay) have logical topological equivalence in physics; this ensures the invariance of unitary polarization operator computation results and AGI sensitivity settings under different path configurations;
- This "roughly arbitrary" guidance paradigm brings three-dimensional dimensionality reduction to engineering implementation, and the software-defined Vref further amplifies the flexibility of the architecture:
- Physical layer: Fault tolerance and dynamic avoidance, capable of bypassing defective "bad points" and adjusting paths dynamically without affecting the logical computation of polarization operators and the setting of AGI sensitivity; the Vref can be real-time adjusted to compensate for the polarization state change caused by path adjustment;
- Compilation layer: From "full routing" to "constraint solving", only the angle combination with the lowest energy consumption and least crosstalk needs to be found for polarization state evolution, and the sensitivity requirement is converted into Vref calibration parameters and integrated into the topological design, realizing the unified compilation of optical layer topology and electrical layer threshold;
- Operation layer: Spatial compression of high-dimensional information, multiple logical skeletons can be written overlapping in the same space to realize multi-engine parallel polarization computing, and each parallel engine is configured with an independent Vref threshold to support differentiated sensitivity requirements of different AGI tasks, further

amplifying the parallelism and flexibility advantage of the polarization scheme.

## 2. Three-Level Self-Healing Control Algorithm

The three-level self-healing control algorithm is designed to maintain the stability of polarization state evolution and the accuracy of unitary conjugate transposition inverse computation, and integrates the software-programmable  $V_{ref}$  calibration module to realize dynamic adjustment of AGI computing sensitivity, ensuring the reliability of spatial topology-based computing and the flexibility of algorithm-physical matching:

### (1) Spatial Addressing Self-Calibration Program: Optical Homing

Establish a dynamic mapping relationship between the 2D pixel coordinates  $(x,y)$  of the SLM and the 3D voxels  $(X,Y,Z)$  inside the cube, ensuring precise implantation of polarization operators at specified voxels; the calibration result is used to initialize the comparator  $V_{ref}$  threshold to match the polarization state distribution of the addressed voxel and lay the foundation for subsequent sensitivity adjustment:

- Blind scanning: Randomly select SLM array points and emit low-power UV+IR detection pulses;
- Centroid inversion: Calculate the physical centroid of the intersection point by the trilateration method through the signal intensity difference of the five-side PD array;
- Error modeling: Establish a deviation function  $E(X,Y,Z)$  between the physical centroid and the SLM command  $(\theta,\varphi)$  to form a calibration table, compensating for addressing errors to ensure the accuracy of unitary polarization operator writing; according to the polarization signal intensity at the addressed voxel, initialize the comparator  $V_{ref}$  to the optimal initial threshold to ensure the signal-to-noise ratio and judgment precision of the initial state.

### (2) Virtual Waveguide Enhancement Program: Self-Organizing Routing

Convert the logical sketch into a "logical manifold" inside the cube and realize path self-organization by utilizing reinforcement learning, optimizing the spatial topology of polarization computing for higher efficiency; the reinforcement learning feedback is fused with the  $V_{ref}$  adjustment signal, and the path optimization and sensitivity calibration are carried out synchronously to match the AGI task requirements:

- Preset path: The SLM emits UV/IR hybrid light at a probability-optimal angle according to the logical sketch, initializing the polarization evolution path; the silicon-based controller sets the initial  $V_{ref}$  threshold according to the AGI task sensitivity requirements corresponding to the logical sketch;
- Multi-directional feedback: If the polarization projection received by the PD is close to the target, increase the IR duty cycle of the corresponding SLM pixel to reinforce the path, optimizing the refractive index gradient for more stable polarization state evolution; if the judgment precision is insufficient or the noise is too high, adjust the  $V_{ref}$  threshold in real time according to the PD feedback signal to balance the computing sensitivity and robustness;

- Physical collapse: Utilize the  $Tm^{3+}$  stimulated saturation effect, the path that first forms a logical closed loop will automatically suppress stray paths through nonlinear gain, forming a logical manifold with the least resistance and highest polarization fidelity; the final stable path is matched with the optimized  $V_{ref}$  threshold, ensuring the efficiency and accuracy of unitary matrix computation, spatial topology-based logical judgment and AGI dynamic sensitivity adjustment.

### (3) Runtime Polarization Correction Program: Vector Lock-loop

Aim at the stability of in-memory computing, correct the polarization deviation caused by thermal drift and refractive index inhomogeneity in real time, maintaining the unitary characteristics of polarization evolution and the accuracy of conjugate transposition inverse; the loop synchronously calibrates the comparator  $V_{ref}$  according to the polarization deviation and AGI task state, realizing the joint correction of polarization state and sensitivity:

- Feature extraction: The PD captures the current polarization rotation angle  $\psi_{real}$  and Stokes parameters, real-time monitoring the polarization state evolution and the signal intensity change of the PD output;
- Deviation comparison: Compare  $\psi_{real}$  with the target  $\psi_{target}$  to calculate the polarization error, judging the deviation of the polarization state from the unitary evolution track; at the same time, judge whether the PD output signal is within the optimal judgment range of the current  $V_{ref}$  threshold, and calculate the sensitivity adjustment demand;
- Dynamic compensation: Without changing the path, only fine-tune the initial polarization angle of the SLM to inject a "reverse polarization deviation" to offset the unintended birefringence inside the medium, realizing active polarization locking; if the PD output signal deviates from the optimal judgment range, synchronously adjust the comparator  $V_{ref}$  to the new optimal threshold according to the deviation degree, ensuring the correctness of unitary conjugate transposition inverse computation, spatial topology-based logical judgment and the consistency of AGI computing sensitivity.

## V. Photonic High-Level Synthesis (Photonics HLS): From High-Level Language to Physical Collapse

### 1. Core of the Paradigm: Connecting Photonic Hardware and Software Definition

Map the mature logical architecture of the silicon-based world to the physical medium of quartz through a set of "optical instruction sets" as a bridge, realizing high-level synthesis in the field of photonic computing, and directly mapping the unitary polarization computing operators of the PDMM architecture to the Optical ISA; the paradigm is further expanded to integrate AGI computing sensitivity adjustment into the high-level synthesis flow, and the software-programmable  $V_{ref}$  of the PD comparator is taken as a core hardware parameter to realize zero-loss compilation from high-level language algorithm requirements to physical layer polarization computing and sensitivity calibration:

- Humans are no longer "circuit repairers" but "architecture designers", focusing on logical abstraction and instruction set definition, and directly calling polarization computing operators such as unitary inverse matrix and AGI sensitivity adjustment instructions according to the algorithm requirements of different AGI tasks;
- The silicon-based system acts as an offline compiler, responsible for topological sketching and linearization, converting high-level logical instructions into polarization evolution paths and spatial topology configurations, and at the same time converting AGI sensitivity requirements into comparator Vref calibration parameters, realizing the unified mapping of algorithm logic and sensitivity parameters to hardware parameters;
- The SLM compiler is responsible for physical layer writing and verification, realizing a complete compilation flow from high-level language to physical collapse, and directly implanting the corresponding polarization operators in the 3D voxels of the quartz cube while synchronously configuring the Vref threshold of the PD comparator according to the compiled parameters, realizing the synchronous implantation of optical layer operators and electrical layer threshold calibration.

## 2. Complete Compilation Division Flow

The complete compilation flow is fully compatible with the polarization incoherent computing paradigm and unitary matrix computing characteristics, and integrates the AGI sensitivity adjustment based on programmable Vref, realizing the direct mapping from high-level language to spatial topology-based polarization computing and dynamic sensitivity calibration:

### (1) Human Layer: Semantic Extraction and Optical Instruction Set Definition

The core layer for logical abstraction, directly compatible with the PDMM architecture's POL/TM/PD series operators, and encapsulating the unitary polarization computing characteristics and AGI dynamic sensitivity adjustment into easy-to-call optical instructions:

- Logical abstraction: Extract large model operators (e.g., Attention, Softmax, MLP) from high-level programming languages, and decompose them into basic polarization computing operations such as unitary matrix transposition, conjugate, and inverse; at the same time, extract the sensitivity requirements of different algorithm modules (e.g., high sensitivity for feature extraction modules, low sensitivity for classification decision modules) according to the AGI task characteristics;
- Instruction translation: Develop a set of Optical Instruction Set Architecture (Optical ISA), such as ROTATE\_POL(P1, 90), VECTOR\_DOT(V1, V2), and directly map the unitary inverse matrix computation to the INV\_POL(M1) optical instruction, the tensor transformation to POL\_TRANS(T1, M1); add the SET\_SENS(Vref, Task\_ID) sensitivity adjustment instruction to map the AGI task sensitivity requirements to the comparator Vref calibration parameters, and support the real-time calling of the instruction during the AGI task execution;
- High-dimensional compression: Utilize the multiple degrees of freedom of light

(wavelength, polarization, spatial position) to compress thousands of lines of silicon-based binary instructions into a single "high-dimensional optical instruction packet"; the Vref calibration parameters are encapsulated into the instruction packet as auxiliary parameters, realizing the synchronous transmission of computing instructions and sensitivity parameters, and amplifying the parallelism advantage of polarization computing.

## (2) Silicon-Based Layer: Topological Sketching and Linearization

The offline compilation layer, converting optical instructions into 3D spatial topology and polarization evolution paths, matching the physical characteristics of the quartz cube, and converting sensitivity adjustment instructions into Vref calibration tables and synchronous control signals:

- Offline compiler: Generate a 3D logical topological sketch in the simulated space according to the optical instruction set, designing the spatial distribution of voxel activation and polarization operator implantation based on the principle of spatial topology replacing temporal phase; at the same time, generate a Vref calibration table according to the SET\_SENS instruction, matching different Vref thresholds for different logical modules and AGI task stages;
- Linear mapping: Simplify the 3D optical field path into a linear mapping of "input surface coordinate → logical voxel point → polarization transformation point → output surface projection", ensuring the smooth evolution of the polarization state and the correctness of unitary matrix computation; map the Vref calibration table to the time sequence and spatial position of the PD comparator, realizing the partitioned and time-sequential calibration of the threshold;
- SLM bitstream generation: Output an executable "phase/intensity/polarization" configuration table, including the parameters of UV/IR cross modulation for voxel activation and polarization operator writing; at the same time, output the Vref calibration bitstream of the PD comparator, including the threshold value, calibration time sequence and partition information, realizing the unified generation of optical layer and electrical layer control parameters.

## (3) SLM Compiler Layer: Physical Layer Writing and Verification

The physical execution layer, responsible for the parallel writing of polarization operators and real-time verification of computing results, with closed-loop correction via the five-side PD array, and synchronously completing the Vref calibration of the comparator and real-time adjustment according to the verification result:

- Parallel execution: The SLM receives the configuration table, emits UV/IR hybrid light, and induces local birefringence in the specified 3D voxels, directly implanting the polarization operators such as unitary conjugate transposition inverse; the silicon-based controller synchronously configures the Vref threshold of the PD comparator according to the Vref calibration bitstream, realizing the synchronous start of optical layer computing and electrical layer judgment with the optimal sensitivity;

- Runtime check: The five-side PD array verifies the polarization projection results and Stokes parameters in real time, monitoring the accuracy of unitary matrix computation and spatial topology-based logical judgment; at the same time, monitor the matching degree between the PD output signal and the current Vref threshold, and feed back the sensitivity adjustment demand to the silicon-based controller if the matching degree is insufficient;
- Dynamic correction: If the projection is inconsistent, the compiler fine-tunes the power or wavefront of the SLM in real time to adjust the polarization state evolution, maintaining the correctness of the logical topology and the unitary characteristics of polarization computing; if the sensitivity matching degree is insufficient, the silicon-based controller adjusts the comparator Vref threshold in real time according to the feedback signal, or calls the SET\_SENS instruction to update the sensitivity parameter, realizing the dynamic optimization of AGI computing sensitivity.

### 3. Mode Selection: Static Curing vs. Streaming Trigger

Two operating modes are supported, both adapting to the polarization incoherent computing paradigm, and the software-programmable Vref of the PD comparator is fully compatible with both modes, realizing the dynamic adjustment of AGI computing sensitivity for different application scenarios:

- Static curing (Programmable Photonic FPGA): Write all polarization logic gates and unitary operators first, then let the signal light pass through; it has low requirements for SLM refresh bandwidth and is suitable for fixed inference tasks with stable computing logic, such as pre-trained large model inference with fixed unitary matrix weights; the comparator Vref threshold is solidified in the silicon-based controller according to the sensitivity requirements of the fixed task, and a small-range real-time calibration is performed only for environmental drift, ensuring the long-term consistency of AGI computing sensitivity;
- Streaming trigger (Dynamic Processor): Adjust the topological structure and polarization operator implantation while computing with high flexibility, suitable for dynamic models and online learning tasks that require real-time adjustment of unitary matrix parameters and spatial topology; the comparator Vref threshold supports real-time high-precision adjustment, and can be synchronously updated with the topological structure adjustment according to the real-time changing sensitivity requirements of the AGI task, realizing the dynamic matching of optical layer topology, unitary matrix parameters and electrical layer sensitivity;
- Compromise solution: Adopt the "static skeleton + dynamic fine-tuning" mode, where core polarization operators (e.g., unitary conjugate transposition inverse, basic tensor transformation) are cured, and only polarization angle fine-tuning is performed during runtime to compensate for deviations; the corresponding Vref adjustment strategy is the "fixed core threshold + dynamic fine-tuning", where the Vref threshold of the core operator matching part is solidified, and the threshold of the fine-tuning part is adjusted in real time according to the AGI task state, balancing performance and flexibility, and



maximizing the parallelism and sensitivity adjustment advantage of the polarization scheme.

## VI. Core Value and Future Research Directions of the New System

### 1. Core Value

The new system builds a complete polarization incoherent computing architecture with spatial topology replacing temporal phase as the core, and adds the software-programmable Vref of PD front-end comparators to realize dynamic adjustment of AGI computing sensitivity; its core value lies in the subversive computing advantages brought by unitary matrix physical computing, 5-channel differential PD noise suppression and software-defined sensitivity, as well as high engineering feasibility:

- Architecture simplification: Single-plane SLM regulation without side pumping, realizing monolithic integration of optical paths, and the geometric characteristics of optical fields realize the decoupling of addressing and computing, fundamentally eliminating path errors; the integration of programmable Vref into the PD array does not increase the hardware complexity of the optical layer, and only adds a simple digital control module to the electrical layer, conforming to the design principle of architecture simplification;
- Robustness improvement: Polarization incoherent computing combined with self-healing control, 5-channel differential PD eliminates intensity superposition common-mode noise, and the programmable Vref can real-time calibrate the judgment threshold to compensate for environmental drift, maintaining high SNR under ambient conditions, breaking away from the dependence on precise temperature control and anti-vibration platforms; the dynamic adjustment of sensitivity further improves the robustness of the system for different AGI task noise characteristics;
- Unprecedented computing efficiency and flexibility: Polarization scheme doubles the initial parallelism compared with single-phase schemes, polarization evolution follows unitary matrix rules with conjugate transposition realizing  $O(1)$  time complexity inverse computation, forming an insurmountable core advantage over traditional phase-interference-based photonic computing architectures; the software-programmable Vref realizes the dynamic adjustment of AGI computing sensitivity, making the hardware layer have the flexibility to adapt to different AGI task requirements, and filling the gap between photonic computing hardware and AGI algorithm dynamic demand;
- In-depth software definition: Photonics HLS paradigm greatly reduces the development threshold, directly mapping PDMM unitary polarization operators to Optical ISA, and integrating AGI sensitivity adjustment into the optical instruction set, realizing zero-loss compilation from high-level language to physical polarization computing and sensitivity calibration; the whole architecture from optical layer topology to electrical layer judgment threshold is software-defined, truly realizing the "algorithm defines hardware" photonic computing mode adapted to AGI;
- Engineering adaptability: The full link reuses mature processes, featuring low cost and mass production feasibility, multiple virtual waveguide writing methods support

scenario-based customization, and are compatible with existing silicon-based EDA tools; the programmable Vref module adopts mature analog-digital hybrid integrated circuit technology, which can be monolithically integrated with the PD array and silicon-based controller without additional process development, further improving the engineering realizability;

- Flexible expansion and multi-task adaptation: Multiple virtual waveguide writing methods support scenario-based customization and combination, adapting to implementation scenarios with different computing power requirements; the logical topological equivalence paradigm supports dynamic path adjustment and multi-engine parallel computing; the partitioned Vref calibration function supports multi-engine parallel computing with different sensitivity requirements on a single chip, realizing the simultaneous execution of multiple AGI tasks with differentiated sensitivity requirements, and greatly improving the multi-task processing capability of the chip.

## 2. Future Research Directions

Focus on the optimization of polarization incoherent computing performance, the improvement of software-defined capabilities, the deep integration of AGI dynamic sensitivity adjustment and algorithm, and engineering prototype verification, further amplifying the advantages of spatial topology replacing temporal phase, unitary matrix physical computing and software-programmable Vref:

- Develop a dedicated optical instruction set for large model operators, optimize the mapping of high-dimensional tensor computing and unitary matrix operations to Optical ISA, and enrich the SET\_SENS instruction set to support multi-dimensional sensitivity adjustment (e.g., spatial partition, time sequence, task type) and automatic matching of sensitivity parameters, improve compilation efficiency and execution density, and adapt to the characteristics of large model parallel computing and multi-task fusion;
- Realize millisecond-level "logical rerouting", optimize the self-organizing routing and vector lock-loop algorithms, improve the dynamic adjustment speed of virtual waveguides and polarization states, and optimize the Vref real-time adjustment algorithm to reduce the response time of sensitivity calibration, endow the quartz cube with the evolutionary ability to adapt to algorithm evolution and real-time computing task changes, and realize the synchronous millisecond-level update of optical layer topology and electrical layer sensitivity;
- Complete the integrated integration of SLM, quartz cube, five-side PD array with programmable Vref and silicon-based controller, build a physical prototype, and conduct comprehensive performance testing and scenario verification (e.g., large model inference, high-dimensional matrix computing, AGI online learning) to verify the  $O(1)$  inverse computation efficiency, parallelism advantage of unitary polarization computing and the effectiveness of dynamic sensitivity adjustment for different AGI tasks;
- Carry out optimization research on the combination of multiple virtual waveguide writing methods and Vref adjustment strategies, establish a writing method-Vref threshold matching system for different AGI task scenarios, optimize the induction efficiency and

polarization sensitivity of virtual waveguides, and improve the stability and reconfigurability of polarization operator implantation and sensitivity calibration;

- Optimize the photosensitive/thermo-optical characteristics of  $\text{Tm}^{3+}$ -doped quartz, adjust the doping concentration and process parameters, improve the induction efficiency and stability of local birefringence/dichroism, and reduce the polarization evolution loss in the 3D lattice to further improve the computing accuracy of unitary matrix operators; at the same time, optimize the PD array and comparator integration process, improve the response speed and precision of the comparator, and expand the adjustable range of  $V_{\text{ref}}$  to adapt to more complex AGI sensitivity requirements;

- Explore the multi-cube cascading expansion scheme, based on the logical topological equivalence paradigm, realize the large-scale expansion of polarization incoherent computing, and develop a unified  $V_{\text{ref}}$  calibration and sensitivity coordination algorithm for the cascaded system, realize the global dynamic sensitivity adjustment and partitioned coordination of the cascaded chip array, and meet the computing power and multi-dimensional sensitivity requirements of ultra-large-scale AGI models and high-performance scientific computing;

- Carry out research on the automatic sensitivity matching algorithm of AGI-photonics computing, realize the real-time extraction of AGI task sensitivity requirements according to the algorithm running state and data characteristics, and the automatic calibration of the comparator  $V_{\text{ref}}$  threshold through the silicon-based controller, realize the end-to-end automatic matching of AGI algorithm and photonic computing hardware sensitivity, and further improve the intelligence level and adaptive ability of the architecture.